Heating load and COP optimizations for a class of generalized irreversible universal steady-flow variable-temperature heat reservoir heat pump cycle model

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Abstract

The heating load and coefficient of performance (COP) of a class of generalized irreversible universal steady-flow heat pump cycle model with variable-temperature heat reservoirs and the losses of heat transfer, heat leakage and internal irreversibility are investigated by using the theory of finite-time thermodynamics. The universal heat pump cycle model consists of two heat-absorbing branches, two heat-releasing branches and two adiabatic branches. Expressions of heating load and COP of the universal heat pump cycle model are deduced, respectively. By means of numerical calculations, the heat conductance distributions between hot- and cold-side heat exchangers are optimized by taking the maximum heating load as objective. There exist both optimal heat conductance distributions and optimal thermal capacity rate matching between the working fluid and heat reservoirs which lead to the double maximum heating load. The effects of heat leakage, internal irreversibility, total heat exchanger inventory and thermal capacity rate of the working fluid on the optimal performance of the cycle are discussed in detail. The results obtained herein include the optimal performance of endoreversible and irreversible, constant- and variable-temperature heat reservoir Brayton, Otto, Diesel, Atkinson, Dual, Miller and Carnot heat pump cycles. The results obtained can provide some theoretical guidelines for the designs and operations of practical heat pumps.

Keywords: finite-time thermodynamics; heating load; COP; generalized irreversible universal heat pump cycle; heat leakage; optimal heat capacity rate matching

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1 INTRODUCTION

The air-standard heat pumps, utilized air as refrigerant, have little emissions of carbon oxides and a high coefficient of performance (COP), and have gotten a broad prospect of development. It is considered that the air-standard heat pumps are friendly to the environment and energy saving, which accord with the viewpoint of low-carbon and sustainable development strategy.

Recently, analyses and optimizations of various heat pump cycles have made tremendous progress by using finite-time thermodynamics [1–11]. Blanchard [12], Goth and Feidt [13] and Feidt [14–16] derived the optimal COP for the fixed heating load for endoreversible Carnot heat pump. Sun and colleagues [17,18] extended the characteristic parameters of heat engines to the heat pumps, and derived the optimization criteria of heat pump cycles between heat reservoirs. Wu [19] and Chen *et al.* [20] investigated the specific heating load and COP performance of an endoreversible Carnot heat pump. Sahin and Kodal [21] investigated the finite-time thermoeconomic performance of an endoreversible Carnot heat pump. Li *et al.* [22] further investigated the heating load versus COP characteristic of an endoreversible Carnot heat pump with complex heat transfer law $q \propto \Delta(T^n)^m$. Wu *et al.* [23] derived the optimal relation between the heating load and COP of endoreversible Carnot and Brayton heat pumps, and made some comparisons between the two cycle performances for the same boundary conditions. Chen *et al.* [24] compared the performance of an endoreversible variable-temperature heat reservoir Brayton heat pump cycle with that of a Carnot heat pump cycle in detail, and found that heating load of the Brayton heat pump cycle is always higher than that of the Carnot heat pump cycle for the same boundary conditions and COP. The best explanation for this was that

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optimal heat capacity rate matching between the working fluid and the heat reservoirs was achieved in the Brayton heat pump cycle. Bi *et al.* [25,26] investigated the optimal performance of endoreversible Brayton heat pumps with constant- and variable-temperature heat reservoirs by taking heating load, COP and heating load density as the optimization objectives. A similar optimization was carried out by taking ecological function and exergetic efficiency as objectives [27,28].

Practical heat pump cycles are always irreversible ones and with variable-temperature heat reservoirs. Refs [29–37] investigated the optimal performance of irreversible Carnot heat pump and the effect of heat transfer law on its performance. Chen *et al.* [38] analyzed the heating load and COP performance of irreversible constant- and variable-temperature heat reservoirs air (Brayton) heat pump cycles. Bi *et al.* [39] investigated the optimal performance of irreversible air heat pump cycle by taking maximum heating load and the maximum COP as objectives, respectively, and discussed the effects of the heat reservoir temperature ratio, the total heat exchanger inventory and the efficiencies of the compressor and expander on the two optimal performances. Bi *et al.* [40] optimized the heating load, exergetic efficiency and ecological criterion for irreversible simple air heat pump cycles with variable-temperature heat reservoirs, and made some performance comparisons among the three optimization objectives.

For variable-temperature heat reservoir heat pumps, studies on irreversibilities mainly concentrate on heat resistance and internal irreversibility [23,24,26,28,30,31,38,40]. But, heat leakage always exists in practical heat pumps. Therefore, heat leakage is considered in this paper, which will further approach the practical heat pump cycles. Moreover, despite many studies on various heat pump cycles, the performance of a universal heat pump cycle with variable-temperature heat reservoirs is rarely studied. One of the aims of finite-time thermodynamics is to pursue generalized rules and results. On the basis of these considerations, this paper will build a generalized irreversible universal steady-flow heat pump cycle model consisting of two heat-absorbing branches, two heat-releasing branches and two adiabatic branches with variable-temperature heat reservoirs and the losses of heat transfer, heat leakage and internal irreversibility. Heating load will be taken as an optimization objective for performance optimization of the universal heat pump cycle model. This paper also discusses a universal heat pump cycle model with variable-temperature heat reservoirs and the losses of heat transfer, heat leakage and internal irreversibility and to study the optimal heating load and COP characteristics of the universal heat pump cycle. This paper first considers heat leakage in the universal heat pump cycle with variable-temperature heat reservoirs, i.e. heat resistance, heat leakage and internal irreversibility are simultaneously integrated in the variable-temperature heat reservoir universal heat pump cycle model, which is a unique contribution of this paper. The results obtained herein include the optimal performance of endoreversible and irreversible, constant- and variable- temperature heat reservoir Brayton, Otto, Diesel, Atkinson, Dual, Miller and Carnot heat pump cycles, which have been investigated in many literatures [12–18,22–34,36–40]. The universal cycle model introduces two variable-temperature heat reservoirs and includes the losses of heat transfer, heat leakage and internal irreversibility, which is more close to practical heat pumps. The results obtained will provide some theoretical guidelines for the designs and operations of practical heat pumps.

2 CYCLE MODEL

The schematic diagram of a generalized irreversible variable-temperature heat reservoir universal heat pump cycle model with heat resistance, heat leakage and internal irreversibility is shown in Figure 1. The T–s diagram of the universal heat pump cycle is shown in Figure 2. The following assumptions are made for this model:

(1) The working fluid is an ideal gas and flows through the system in a quasi-steady fashion. The cycle consists of two heat-absorbing branches (1–2 and 2–3) with constant working fluid thermal capacity rates (mass flow rate of the working fluid and specific heat product) C_{wf1} and C_{wf2}, two heat-releasing branches (4–5 and 5–6) with constant working fluid thermal capacity rates C_{wf4} and C_{wf3} and two adiabatic branches (3–4 and 6–1). All six processes are irreversible.

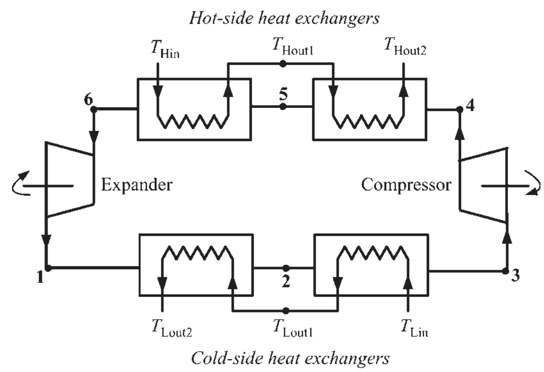

Figure 1. *Schematic diagram of a generalized irreversible variable-temperature heat reservoir universal heat pump cycle model.*

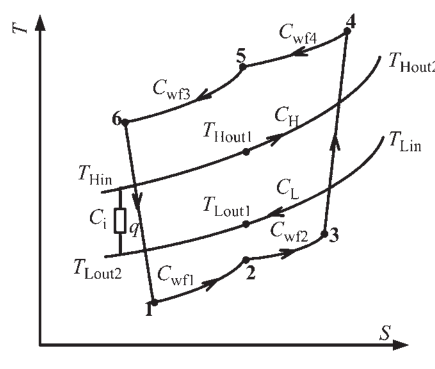

Figure 2. T–s *diagram of the cycle model.*

(2) The hot- and cold-side heat exchangers are considered to be counter-flow heat exchangers, the working fluid temperatures are different from the heat reservoir temperatures owing to the heat transfer. The high-temperature (hot-side) heat sink is considered with a thermal capacity rate C_H, and the inlet and outlet temperatures of the cooling fluid are T_{Hin}, T_{Hout1} and T_{Hout2}, respectively. The low-temperature (cold-side) heat source is considered with a thermal capacity rate C_L, and the inlet and outlet temperatures of the heating fluid are T_{Lin}, T_{Lout1} and T_{Lout2}, respectively.

(3) There exists a bypass heat leakage (q) from the heat sink to the heat source. This bypass heat leakage model was first advanced by Bejan [41,42] for the Carnot heat engine and refrigerator and by Chen et al. [43] for the Carnot heat pump. Thus:

$$q = C_i \Delta T_m = \frac{C_i(T_{Hout2} - T_{Lin}) - (T_{Hin} - T_{Lout2})}{\ln(T_{Hout2} - T_{Lin})/(T_{Hin} - T_{Lout2})} \quad (1)$$

$$Q_H = Q_{H1} + Q_{H2} - q, \quad Q_L = Q_{L1} + Q_{L2} - q \quad (2)$$

where C_i is the heat leakage coefficient, ΔT_m denotes the logarithm average temperature differences of the heat reservoirs. $Q_{H1} + Q_{H2}$ is due to the driving force of temperature differences between the high-temperature heat sink and the working fluid; $Q_{L1} + Q_{L2}$ is due to the driving force of temperature differences between the low-temperature heat source and the working fluid; Q_H is the rate of heat transfer released to the heat sink, i.e. the heating load of the cycle and Q_L the rate of heat transfer supplied by the heat source.

(4) A constant coefficient ϕ is introduced to characterize the additional internal miscellaneous irreversibility effects: $\phi = (Q_{H1} + Q_{H2})/(Q'_{H1} + Q'_{H2}) \geq 1$, where $Q_{H1} + Q_{H2}$ is the rate of heat flow from the warm working fluid to the heat sink for the irreversible cycle model, and $Q_{H1}' + Q_{H2}'$ is that for the endoreversible cycle model with the only loss of heat resistance.

To summarize, the generalized irreversible universal heat pump cycle model with variable-temperature heat reservoirs is characterized by the following three aspects:

(1) The different values of C_H and C_L. If $C_H \to \infty$ and $C_L \to \infty$, the cycle model can be reduced to a generalized irreversible universal heat pump cycle model with constant-temperature heat reservoirs.

(2) The different values of C_{wf1}, C_{wf2}, C_{wf3} and C_{wf4}. If C_{wf1}, C_{wf2}, C_{wf3} and C_{wf4} have different values, the cycle model can be reduced to various special heat pump cycles.

(3) The different values of C_i and ϕ. If $q = 0$ and $\phi = 1$, the model is reduced to an endoreversible universal heat pump cycle model with the only loss of heat resistance. If $q = 0$ and $\phi > 1$, the model is reduced to an irreversible universal heat pump cycle model with heat resistance and internal irreversibility. If $q > 0$ and $\phi = 1$, the model is reduced to an irreversible universal heat pump cycle model with heat resistance and heat-leakage losses.

According to the properties of heat transfer, working fluid, heat reservoir and the theory of heat exchangers, the heat transfer rates (Q_{H1} and Q_{H2}) released to the heat sink and the heat transfer rates (Q_{L1} and Q_{L2}) supplied by the heat source are, respectively, given by

$$\begin{aligned} Q_{H1} &= \frac{U_{H1}[(T_5 - T_{Hout1}) - (T_6 - T_{Hin})]}{\ln[(T_5 - T_{Hout1})/(T_6 - T_{Hin})]} \\ &= C_H(T_{Hout1} - T_{Hin}) = C_{wf3}(T_5 - T_6) \\ &= C_{H1\,min} E_{H1}(T_5 - T_{Hin}) \end{aligned} \quad (3)$$

$$\begin{aligned} Q_{H2} &= \frac{U_{H2}[(T_4 - T_{Hout2}) - (T_5 - T_{Hout1})]}{\ln[(T_4 - T_{Hout2})/(T_5 - T_{Hout1})]} \\ &= C_H(T_{Hout2} - T_{Hout1}) = C_{wf4}(T_4 - T_5) \\ &= C_{H2\,min} E_{H2}(T_4 - T_{Hout1}) \end{aligned} \quad (4)$$

$$\begin{aligned} Q_{L1} &= \frac{U_{L1}[(T_{Lout1} - T_2) - (T_{Lout2} - T_1)]}{\ln[(T_{Lout1} - T_2)/(T_{Lout2} - T_1)]} \\ &= C_L(T_{Lout1} - T_{Lout2}) = C_{wf1}(T_2 - T_1) \\ &= C_{L1\,min} E_{L1}(T_{Lout1} - T_1) \end{aligned} \quad (5)$$

$$\begin{aligned} Q_{L2} &= \frac{U_{L2}[(T_{Lin} - T_3) - (T_{Lout1} - T_2)]}{\ln[(T_{Lin} - T_3)/(T_{Lout1} - T_2)]} \\ &= C_L(T_{Lin} - T_{Lout1}) = C_{wf2}(T_3 - T_2) \\ &= C_{L2\,min} E_{L2}(T_{Lin} - T_2) \end{aligned} \quad (6)$$

where E_{H1}, E_{H2}, E_{L1} and E_{L2} are the effectivenesses of the hot- and cold-side heat exchangers, and are defined as:

$$E_{H1} = \frac{1 - \exp[-N_{H1}(1 - C_{H1\,min}/C_{H1\,max})]}{1 - (C_{H1\,min}/C_{H1\,max})\exp[-N_{H1}(1 - C_{H1\,min}/C_{H1\,max})]} \quad (7)$$

$$E_{H2} = \frac{1 - \exp[-N_{H2}(1 - C_{H2\,min}/C_{H2\,max})]}{1 - (C_{H2\,min}/C_{H2\,max})\exp[-N_{H2}(1 - C_{H2\,min}/C_{H2\,max})]} \quad (8)$$

$$E_{L1} = \frac{1 - \exp[-N_{L1}(1 - C_{L1\,min}/C_{L1\,max})]}{1 - (C_{L1\,min}/C_{L1\,max})\exp[-N_{L1}(1 - C_{L1\,min}/C_{L1\,max})]} \quad (9)$$

$$E_{L2} = \frac{1 - \exp[-N_{L2}(1 - C_{L2\,min}/C_{L2\,max})]}{1 - (C_{L2\,min}/C_{L2\,max})\exp[-N_{L2}(1 - C_{L2\,min}/C_{L2\,max})]} \quad (10)$$

where C_{H1min} and C_{H1max} are the minimum and maximum of C_H and C_{wf3}, respectively; C_{H2min} and C_{H2max} are the minimum and maximum of C_H and C_{wf4}, respectively; C_{L1min} and C_{L1max} are the minimum and maximum of C_L and C_{wf1}, respectively; C_{L2min} and C_{L2max} are the minimum and maximum of C_L and C_{wf2}, respectively; and N_{H1}, N_{H2}, N_{L1} and N_{L2} are the numbers of heat transfer units of the hot- and cold-side heat exchangers, respectively:

$$C_{H1\,min} = \min\{C_H, C_{wf3}\}, \quad C_{H1\,max} = \max\{C_H, C_{wf3}\} \tag{11}$$

$$C_{H2\,min} = \min\{C_H, C_{wf4}\}, \quad C_{H2\,max} = \max\{C_H, C_{wf4}\} \tag{12}$$

$$C_{L1\,min} = \min\{C_L, C_{wf1}\}, \quad C_{L1\,max} = \max\{C_L, C_{wf1}\} \tag{13}$$

$$C_{L2\,min} = \min\{C_L, C_{wf2}\}, \quad C_{L2\,max} = \max\{C_L, C_{wf2}\} \tag{14}$$

$$N_{H1} = \frac{U_{H1}}{C_{H1\,min}},\ N_{H2} = \frac{U_{H2}}{C_{H2\,min}},\ N_{L1} = \frac{U_{L1}}{C_{L1\,min}},\ N_{L2} = \frac{U_{L2}}{C_{L2\,min}} \tag{15}$$

where U_{H1}, U_{H2}, U_{L1} and U_{L2} are the heat conductances, i.e. the product of heat transfer coefficient α and heat transfer surface area F.

3 PERFORMANCE ANALYSIS

Combining Equations (3–6), one can obtain:

$$T_5 = \frac{C_{wf3}T_6 - C_{H1\,min}E_{H1}T_{Hin}}{C_{wf3} - C_{H1\,min}E_{H1}} \tag{16}$$

$$\begin{aligned} T_4 = {} & [C_{H1\,min}C_{H2\,min}C_{wf3}E_{H1}E_{H2}(-T_6 + T_{Hin})/C_H \\ & + C_{H1\,min}E_{H1}T_{Hin}(-C_{wf4} + C_{H2\,min}E_{H2}) \\ & + C_{wf3}(C_{wf4}T_6 - C_{H2\,min}E_{H2}T_{Hin})]/[(C_{wf3} \\ & - C_{H1\,min}E_{H1})(C_{wf4} - C_{H2\,min}E_{H2})] \end{aligned} \tag{17}$$

$$\begin{aligned} T_1 = {} & [C_{L1\,min}C_{L2\,min}C_{wf2}E_{L1}E_{L2}(T_{Lin} - T_3)/C_L \\ & + C_{L1\,min}E_{L1}T_{Lin}(C_{L2\,min}E_{L2} - C_{wf2}) \\ & + C_{wf1}(C_{wf2}T_3 - C_{L2\,min}E_{L2}T_{Lin})]/[(C_{wf1} \\ & - C_{L1min}E_{L1})(C_{wf2} - C_{L2\,min}E_{L2})] \end{aligned} \tag{18}$$

$$T_2 = \frac{C_{wf2}T_3 - C_{L2\,min}E_{L2}T_{Lin}}{C_{wf2} - C_{L2\,min}E_{L2}} \tag{19}$$

The second law of thermodynamics requires that:

$$\phi = \frac{Q_{H1} + Q_{H2}}{Q'_{H1} + Q'_{H2}} = \frac{C_{wf3}\ln(T_5/T_6) + C_{wf4}\ln(T_4/T_5)}{C_{wf1}\ln(T_2/T_1) + C_{wf2}\ln(T_3/T_2)} \tag{20}$$

Thus:

$$T_2 = T_1 G \tag{21}$$

where:

$$\begin{aligned} G = {} & x^{C_{wf3}/(\phi C_{wf1})} y^{-C_{wf2}/C_{wf1}} \\ & \{[C_{H1\,min}C_{H2\,min}C_{wf3}E_{H1}E_{H2}(-T_6 + T_{Hin})/C_H \\ & + C_{H1\,min}E_{H1}T_{Hin}(-C_{wf4} + C_{H2min}E_{H2}) \\ & + C_{wf3}(C_{wf4}T_6 - C_{H2\,min}E_{H2}T_{Hin})]/ \\ & [(C_{wf3}T_6 - C_{H1\,min}E_{H1}T_{Hin}) \\ & \times(C_{wf2} - C_{H2\,min}E_{H2})]\}^{C_{wf4}/(\phi C_{wf1})} \end{aligned} \tag{22}$$

where $x = T_5/T_6$ and $y = T_3/T_2$.

Combining Equations (3–6) with (18–22) gives:

$$\begin{aligned} T_1 = {} & (C_{L1\,min}E_{L1}T_{Lin}[C_{L2\,min}C_{wf2}E_{L2} \\ & + C_L(-C_{wf2} + C_{L2\,min}E_{L2})])/(GyC_{L1\,min}C_{L2\,min}C_{wf2}E_{L1}E_{L2} \\ & - C_L[C_{wf1}(G-1) + C_{L1\,min}E_{L1}](C_{wf2} - C_{L2\,min}E_{L2})) \end{aligned} \tag{23}$$

$$\begin{aligned} T_2 = {} & (GC_{L1\,min}E_{L1}T_{Lin}[C_{L2\,min}C_{wf2}E_{L2} \\ & + C_L(-C_{wf2} + C_{L2\,min}E_{L2})])/(GyC_{L1\,min}C_{L2\,min}C_{wf2}E_{L1}E_{L2} \\ & - C_L[C_{wf1}(G-1) + C_{L1\,min}E_{L1}](C_{wf2} - C_{L2\,min}E_{L2})) \end{aligned} \tag{24}$$

$$\begin{aligned} T_3 = {} & (GyC_{L1\,min}E_{L1}T_{Lin}[C_{L2\,min}C_{wf2}E_{L2} \\ & + C_L(-C_{wf2} + C_{L2\,min}E_{L2})])/(GyC_{L1\,min}C_{L2\,min}C_{wf2}E_{L1}E_{L2} \\ & - C_L[C_{wf1}(G-1) + C_{L1\,min}E_{L1}](C_{wf2} - C_{L2min}E_{L2})) \end{aligned} \tag{25}$$

$$\begin{aligned} T_{Hout2} = {} & [C_H^2 T_{Hin}(C_{wf3} - C_{H1\,min}E_{H1})(C_{wf4} - C_{H2\,min}E_{H2}) \\ & + C_{H1\,min}C_{H2\,min}C_{wf3}C_{wf4}E_{H1}E_{H2}(-T_6 + T_{Hin}) \\ & + C_H C_{wf3}(T_6 - T_{Hin})(C_{H2\,min}C_{wf4}E_{H2} \\ & + C_{H1\,min}E_{H1}C_{wf4} - C_{H1\,min}C_{H2\,min}E_{H1}E_{H2})]/ \\ & [C_H^2(C_{wf3} - C_{H1\,min}E_{H1})(C_{wf4} - C_{H2\,min}E_{H2})] \end{aligned} \tag{26}$$

$$\begin{aligned} T_{Lout2} = {} & \{C_{L1min}C_{L2min}C_{wf1}C_{wf2}E_{L1}E_{L2}T_{Lin}[(G-1)C_{wf1} \\ & + (1-Gy)C_{L1min}E_{L1}] + C_L^2 T_{Lin}(C_{wf1} - C_{L1min}E_{L1}) \\ & \times(-C_{wf1} + C_{wf1}G + C_{L1min}E_{L1})(C_{wf2} - C_{L2min}E_{L2}) \\ & + C_L C_{wf2}T_{Lin}[(1-G)C_{L2min}C_{wf1}^2E_{L2} + C_{L1min}^2E_{L1}^2 \\ & \times(-C_{wf1} + C_{wf1}Gy + C_{L2min}E_{L2}) + (1-G)C_{wf1} \\ & \times C_{L1min}C_{wf1}E_{L1} + (G-2)C_{L2min}E_{L2}C_{L1min}C_{wf1}E_{L1}]\}/ \\ & (C_{wf1}C_L - C_{L1min}C_LE_{L1})[-GyC_{L1min}C_{L2min}C_{wf2}E_{L1}E_{L2} \\ & + C_L(-C_{wf1} + C_{wf1}G + C_{L1min}E_{L1})(C_{wf2} - C_{L2min}E_{L2})]\} \end{aligned} \tag{27}$$

Substituting Equations (1), (3), (4), (16), (17) into (2) yields the heating load of the cycle:

$$
\begin{aligned}
Q_H &= Q_{H1} + Q_{H2} - q \\
&= [-C_{H1\min}C_{H2\min}C_{wf3}C_{wf4}E_{H1}E_{H2}(T_6 - T_{Hin})/C_H \\
&\quad + C_{H2\min}C_{wf3}C_{wf4}E_{H2}(T_6 - T_{Hin}) \\
&\quad + C_{H1\min}E_{H1}C_{wf3}(T_6 - T_{Hin})(C_{wf4} - C_{H2\min}E_{H2})]/ \\
&\quad [(C_{wf3} - C_{H1\min}E_{H1})(C_{wf4} - C_{H2\min}E_{H2})] \\
&\quad - C_i[(T_{Hout2} - T_{Lin}) - (T_{Hin} - T_{Lout2})]/ \\
&\quad \ln[(T_{Hout2} - T_{Lin})/(T_{Hin} - T_{Lout2})]
\end{aligned} \tag{28}
$$

Substituting Equations (1), (5), (6), (18), (19) into (2) yields the heat transfer rate supplied by the heat source:

$$
\begin{aligned}
Q_L &= Q_{L1} + Q_{L2} - q \\
&= T_{Lin}\{C_{L2\min}^2E_{L2}^2(1 - C_{L1\min}E_{L1}/C_L)[-G(-1 + y) \\
&\quad \times C_{L1\min}C_{wf2}E_{L1}/C_L - (-1 + G)(C_{wf1} - C_{L1\min}E_{L1})] \\
&\quad + C_{L2\min}E_{L2}[C_{L1\min}C_{wf2}E_{L1}(-C_{wf1}G \\
&\quad + C_{wf1} + C_{L1\min}E_{L1}G - C_{L1\min}E_{L1}Gy)/C_L \\
&\quad + (-1 + G)(C_{wf1}C_{wf2} - C_{wf1}C_{L1\min}E_{L1} - C_{wf2}C_{L1\min}E_{L1})] \\
&\quad + (-1 + G)C_{L1\min}C_{wf1}C_{wf2}E_{L1}\}/ \\
&\quad \{[-GyC_{L1\min}C_{L2\min}C_{wf2}E_{L1}E_{L2}/C_L \\
&\quad + (-C_{wf1} + C_{wf1}G + C_{L1\min}E_{L1})(C_{wf2} - C_{L2\min}E_{L2})]\} \\
&\quad - C_i[(T_{Hout2} - T_{Lin}) - (T_{Hin} - T_{Lout2})]/ \\
&\quad \ln[(T_{Hout2} - T_{Lin})/(T_{Hin} - T_{Lout2})]
\end{aligned} \tag{29}
$$

Combining Equations (28) with (29) gives the COP of the cycle:

$$
\begin{aligned}
\beta &= \frac{Q_H}{Q_H - Q_L} \\
&= [-C_{H1\min}C_{H2\min}C_{wf3}C_{wf4}E_{H1}E_{H2}(T_6 - T_{Hin})/C_H \\
&\quad + C_{H2\min}C_{wf3}C_{wf4}E_{H2}(T_6 - T_{Hin}) + C_{H1\min}E_{H1}C_{wf3}(T_6 - T_{Hin}) \\
&\quad \times (C_{wf4} - C_{H2\min}E_{H2})]/[(C_{wf3} - C_{H1\min}E_{H1})(C_{wf4} - C_{H2\min}E_{H2})] \\
&\quad - C_i[(T_{Hout2} - T_{Lin}) - (T_{Hin} - T_{Lout2})]/\ln[(T_{Hout2} - T_{Lin})/ \\
&\quad (T_{Hin} - T_{Lout2})]/\{[-C_{H1\min}C_{H2\min}C_{wf3}C_{wf4} \\
&\quad \times E_{H1}E_{H2}(T_6 - T_{Hin})/C_H + C_{H2\min}C_{wf3}C_{wf4}E_{H2}(T_6 - T_{Hin}) \\
&\quad + C_{H1\min}E_{H1}C_{wf3}(T_6 - T_{Hin})(C_{wf4} - C_{H2\min}E_{H2})]/ \\
&\quad [(C_{wf3} - C_{H1\min}E_{H1})(C_{wf4} - C_{H2\min}E_{H2})] \\
&\quad - T_{Lin}\{C_{L2\min}^2E_{L2}^2(1 - C_{L1\min}E_{L1}/C_L)[-G(-1 + y) \\
&\quad \times C_{L1\min}C_{wf2}E_{L1}/C_L - (-1 + G)(C_{wf1} - C_{L1\min}E_{L1})] \\
&\quad + C_{L2\min}E_{L2}[C_{L1\min}C_{wf2}E_{L1}(-C_{wf1}G \\
&\quad + C_{wf1} + C_{L1\min}E_{L1}G - C_{L1\min}E_{L1}Gy)/C_L \\
&\quad + (-1 + G)(C_{wf1}C_{wf2} - C_{wf1}C_{L1\min}E_{L1} - C_{wf2}C_{L1\min}E_{L1})] \\
&\quad + (-1 + G)C_{L1\min}C_{wf1}C_{wf2}E_{L1}\}/ \\
&\quad \{[-GyC_{L1\min}C_{L2\min}C_{wf2}E_{L1}E_{L2}/C_L \\
&\quad + (-C_{wf1} + C_{wf1}G + C_{L1\min}E_{L1})(C_{wf2} - C_{L2\min}E_{L2})]\}\}
\end{aligned} \tag{30}
$$

where T_{Hout2} and T_{Lout2} are calculated by Equations (26) and (27).

In order to make the cycle operate normally, state point 2 must be between state points 1 and 3, and state point 5 must be between state points 4 and 6. Therefore, the ranges of x and y are:

$$
\begin{aligned}
1 \le x \le\ & [C_{H1\min}C_{H2\min}C_{wf3}E_{H1}E_{H2}(-T_6 + T_{Hin})/C_H \\
& + C_{H1\min}E_{H1}T_{Hin}(-C_{wf4} + C_{H2\min}E_{H2}) \\
& + C_{wf3}(C_{wf4}T_6 - C_{H2\min}E_{H2}T_{Hin})]/ \\
& [T_6(C_{wf3} - C_{H1\min}E_{H1})(C_{wf4} - C_{H2\min}E_{H2})]
\end{aligned} \tag{31}
$$

$$
\begin{aligned}
1 \le y \le\ & x^{C_{wf3}/(\phi C_{wf2})}\{[C_{H1\min}C_{H2\min}C_{wf3}E_{H1}E_{H2}(-T_6 + T_{Hin})/C_H \\
& + C_{H1\min}E_{H1}T_{Hin}(-C_{wf4} + C_{H2\min}E_{H2}) \\
& + C_{wf3}(C_{wf4}T_6 - C_{H2\min}E_{H2}T_{Hin})]/ \\
& [(C_{wf3}T_6 - C_{H1\min}E_{H1}T_{Hin}) \\
& (C_{wf2} - C_{H2\min}E_{H2})]\}^{C_{wf4}/(\phi C_{wf2})}
\end{aligned} \tag{32}
$$

4 DISCUSSION

Equations (28) and (30) are generalized. If C_H, C_L, C_i and ϕ have different values, Equations (28) and (30) can be simplified into the corresponding analytical formulae for various endoreversible and irreversible, constant- and variable-temperature heat reservoir heat pump cycles.

Figure 3 shows the performance characteristics of the generalized irreversible universal heat pump cycle with variable-temperature heat reservoirs. Heat conductances of the hot- and cold-side heat exchangers are set as $U_{H1} = 0$, $U_{L2} = 0$ and $U_{H2} = U_{L1} = 3$ kW/K for the Brayton, Otto, Diesel and Atkinson heat pump cycles; $U_{H1} = U_{H2} = U_{L1} = 2$ kW/K and $U_{L2} = 0$ for Dual heat pump cycle; $U_{H1} = 0$ and $U_{H2} = U_{L1} = U_{L2} = 2$ kW/K for the Miller heat pump cycle, respectively. Heat leakage and internal irreversibility are set as $C_i = 0.005$ kW/K and $\phi = 1.01$, respectively. It shows that the curve of COP versus heating load is a parabolic-like one for each heat pump cycle, i.e. there exist a maximum COP and the corresponding heating load. The theoretical analysis results for real irreversible single-stage [44,45] and two-stage [46,47] thermoelectric heat pump as well as experimental result of a single-stage thermoelectric heat pump [48] showed that the curve of COP versus heating load was a parabolic-like one, and there existed a maximum COP with the corresponding heating load. Therefore, the theoretical analysis results of this paper are consistent with those of practical heat pumps. One can continue to discuss the special cases of the universal heat pump cycle for different thermal capacity rates of the working fluid (C_{wf1}, C_{wf2}, C_{wf3} and C_{wf4}) in detail, whose heating load versus COP curves are also shown in Figure 3.

(1) When $C_{wf1} = C_{wf2} = \dot{m}C_p$ (mass flow rate $\dot{m}$ of the working fluid and constant pressure specific heat C_p product) and $C_{wf3} = C_{wf4} = \dot{m}C_p$, $U_{H1} = 0$, $U_{L2} = 0$ and $x = y = 1$,

Equations (28) and (30) become the performance characteristics of a generalized irreversible variable-temperature heat reservoir steady-flow Brayton heat pump cycle with the losses of heat resistance, heat leakage and internal irreversibility. Specially, if $C_i = 0$ further, the performance characteristics of the generalized irreversible Brayton heat pump cycle with variable-temperature heat reservoirs become the performance characteristics of the endoreversible [23,24,26,28] and irreversible [38,40] Brayton heat pump cycles with variable-temperature heat reservoirs. If $C_i = 0$, $C_H \to \infty$ and $C_L \to \infty$ further, the performance characteristics of the generalized irreversible Brayton heat pump cycle with variable-temperature heat reservoirs become the performance characteristics of the endoreversible [23,25,27] and irreversible [38,39] Brayton heat pump cycles with constant-temperature heat reservoirs.

(2) When $C_{wf1} = C_{wf2} = \dot{m}C_v$ (mass flow rate $\dot{m}$ of the working fluid and constant volume specific heat C_v product) and $C_{wf3} = C_{wf4} = \dot{m}C_v$, $U_{H1} = 0$, $U_{L2} = 0$ and $x = y = 1$, Equations (28) and (30) become the performance characteristics of a generalized irreversible variable-temperature heat reservoir steady-flow Otto heat pump cycle with the losses of heat resistance, heat leakage and internal irreversibility.

(3) When $C_{wf1} = C_{wf2} = \dot{m}C_v$ and $C_{wf3} = C_{wf4} = \dot{m}C_p$, $U_{H1} = 0$, $U_{L2} = 0$ and $x = y = 1$, Equations (28) and (30) become the performance characteristics of a generalized irreversible variable-temperature heat reservoir steady-flow Diesel heat pump cycle with the losses of heat resistance, heat leakage and internal irreversibility.

(4) When $C_{wf1} = C_{wf2} = \dot{m}C_p$ and $C_{wf3} = C_{wf4} = \dot{m}C_v$, $U_{H1} = 0$, $U_{L2} = 0$ and $x = y = 1$, Equations (28) and (30) become the performance characteristics of a generalized irreversible variable-temperature heat reservoir steady-flow Atkinson heat pump cycle with the losses of heat resistance, heat leakage and internal irreversibility.

(5) When $C_{wf1} = C_{wf2} = \dot{m}C_v$, $C_{wf3} = \dot{m}C_v$ and $C_{wf4} = \dot{m}C_p$, $U_{H1} \neq 0$, $U_{H2} \neq 0$, $U_{L2} = 0$ and $y = 1$, Equations (28) and (30) become the performance characteristics of a generalized irreversible variable-temperature heat reservoir steady-flow Dual heat pump cycle with the losses of heat resistance, heat leakage and internal irreversibility. If $U_{H1} = 0$, $U_{L2} = 0$ and $x = y = 1$ further, the Dual heat pump cycle becomes the Diesel heat pump cycle. If $U_{H2} = 0$, $U_{L2} = 0$ and $y = 1$ further, the Dual heat pump cycle becomes the Otto heat pump cycle.

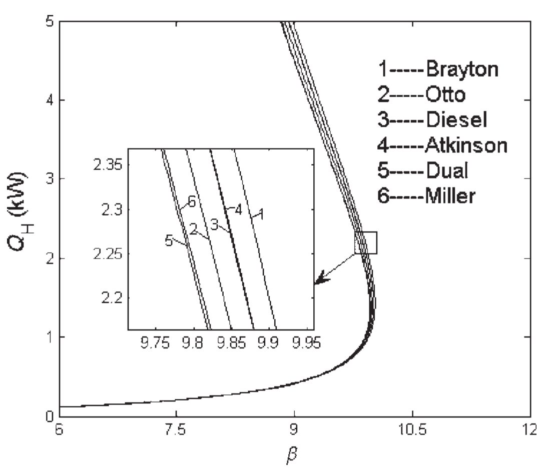

Figure 3. *Heating load versus COP characteristic of an irreversible universal heat pump cycle with variable-temperature heat reservoirs.*

In this case, the range of x becomes:

$$\begin{aligned}1 \le x \le {} & [C_{H1\min}C_{H2\min}C_{wf3}E_{H1}E_{H2}(-T_6 + T_{Hin})/C_H \\ & + C_{H1\min}E_{H1}T_{Hin}(-C_{wf4} + C_{H2\min}E_{H2}) \\ & + C_{wf3}(C_{wf4}T_6 - C_{H2\min}E_{H2}T_{Hin})]/ \\ & [T_6(C_{wf3} - C_{H1\min}E_{H1})(C_{wf4} - C_{H2\min}E_{H2})]\end{aligned} \tag{33}$$

and the value of x is given by:

$$\begin{aligned}x = T_5/T_6 = {} & [C_{H1\min}C_{H2\min}C_{wf3}E_{H1}E_{H2}(-T_6 + T_{Hin})/C_H \\ & + C_{H1\min}E_{H1}T_{Hin}(-C_{wf4} + C_{H2\min}E_{H2}) \\ & + C_{wf3}(C_{wf4}T_6 - C_{H2\min}E_{H2}T_{Hin})]/ \\ & [(C_{wf3}T_6 - C_{H1\min}E_{H1}T_{Hin})(C_{wf2} - C_{H2\min}E_{H2})]\end{aligned} \tag{34}$$

(6) When $C_{wf1} = \dot{m}C_p$, $C_{wf2} = \dot{m}C_v$ and $C_{wf3} = C_{wf4} = \dot{m}C_v$, $U_{H1} = 0$, $U_{L1} \neq 0$, $U_{L2} \neq 0$ and $x = 1$, Equations (28) and (30) become the performance characteristics of a generalized irreversible variable-temperature heat reservoir steady-flow Miller heat pump cycle with the losses of heat resistance, heat leakage and internal irreversibility. If $U_{H1} = 0$, $U_{L2} = 0$ and $x = y = 1$ further, the Miller heat pump cycle becomes the Atkinson heat pump cycle. If $U_{H1} = 0$, $U_{L1} = 0$ and $x = 1$ further, the Miller heat pump cycle becomes the Otto heat pump cycle.

In this case, the range of y is:

$$\begin{aligned}1 \le y \le {} & \{[C_{H1\min}C_{H2\min}C_{wf3}E_{H1}E_{H2}(-T_6 + T_{Hin})/C_H \\ & + C_{H1\min}E_{H1}T_{Hin}(-C_{wf4} + C_{H2\min}E_{H2}) \\ & + C_{wf3}(C_{wf4}T_6 - C_{H2\min}E_{H2}T_{Hin})]/ \\ & [(C_{wf3}T_6 - C_{H1\min}E_{H1}T_{Hin})(C_{wf2} - C_{H2\min}E_{H2})]\}^{1/\phi}\end{aligned} \tag{35}$$

Combining Equations (18), (19) and (22) gives the following equation that the working fluid temperature T_3 should satisfy:

$$\begin{aligned}& (C_{wf1} - C_{L1\min}E_{L1})(C_{wf2}T_3 - C_{L2\min}E_{L2}T_{Lin}) \\ & \quad [T_3(C_{wf2} - C_{L2\min}E_{L2})/(C_{wf2}T_3 - C_{L2\min}E_{L2}T_{Lin})]^{1/k}/ \\ & \quad [C_{L1\min}C_{L2\min}C_{wf2}E_{L1}E_{L2}(T_{Lin} - T_3)/C_L \\ & \quad + C_{L1\min}E_{L1}T_{Lin}(C_{L2\min}E_{L2} - C_{wf2}) \\ & \quad + C_{wf1}(C_{wf2}T_3 - C_{L2\min}E_{L2}T_{Lin})] \\ & = \{[C_{H1\min}C_{H2\min}C_{wf3}E_{H1}E_{H2}(-T_6 + T_{Hin})/C_H \\ & \quad + C_{H1\min}E_{H1}T_{Hin}(-C_{wf4} + C_{H2\min}E_{H2}) \\ & \quad + C_{wf3}(C_{wf4}T_6 - C_{H2\min}E_{H2}T_{Hin})]/ \\ & \quad [(C_{wf3}T_6 - C_{H1\min}E_{H1}T_{Hin})(C_{wf2} - C_{H2\min}E_{H2})]\}^{1/(\phi k)}\end{aligned} \tag{36}$$

where k is the ratio of the specific heats. Moreover, combining Equations (18), (19), (21) with (36) gives G and y.

(7) When $C_{wf1} = C_{wf2} = C_{wf3} = C_{wf4} \to \infty$, Equations (28) and (30) become the performance characteristics of a generalized irreversible variable-temperature heat reservoir steady-flow Carnot heat pump cycle with the losses of heat resistance, heat leakage and internal irreversibility. Specially, if $C_i = 0$ and $\phi = 1.0$ further, the performance characteristics of the generalized irreversible Carnot heat pump cycle with variable-temperature heat reservoirs become the performance characteristics of the endoreversible Carnot heat pump cycle with variable-temperature heat reservoirs [23,24]. If $C_i = 0$, $C_H \to \infty$ and $C_L \to \infty$ further, the performance characteristics of the generalized irreversible Carnot heat pump cycle with variable-temperature heat reservoirs become the performance characteristics of the endoreversible [12–23] and irreversible [29–37] Carnot heat pump cycles with constant-temperature heat reservoirs.

5 PERFORMANCE OPTIMIZATION

5.1 Optimal distributions of heat conductance

If heat conductances of hot- and cold-side heat exchangers are changeable, the heating load of the universal heat pump cycle may be optimized by searching the optimal heat conductance distributions for the fixed total heat exchanger inventory. For the fixed heat exchanger inventory U_T, that is, for the constraint of $U_{H1} + U_{H2} + U_{L1} + U_{L2} = U_T$, defining the distributions of heat conductance $u_{H1} = U_{H1}/U_T$, $u_{H2} = U_{H2}/U_T$, $u_{L1} = U_{L1}/U_T$ and $u_{L2} = U_{L2}/U_T$ leads to:

$$U_{H1} = u_{H1}U_T, \quad U_{H2} = u_{H2}U_T, \quad U_{L1} = u_{L1}U_T, \quad U_{L1} = u_{L1}U_T, \quad U_{L2} = u_{L2}U_T \tag{37}$$

The following conditions should be satisfied: $0 \le u_{H1} \le 1$, $0 \le u_{H2} \le 1$, $0 \le u_{L1} \le 1$, $0 \le u_{L2} \le 1$ and $u_{H1} + u_{H2} + u_{L1} + u_{L2} = 1$. Moreover, heat conductance distributions are set as $u_{H1} = 0$ and $u_{L2} = 0$ for the Brayton, Otto, Diesel and Atkinson heat pump cycles; $u_{L2} = 0$ for the Dual heat pump cycle; $u_{H1} = 0$ for the Miller heat pump cycle, respectively. To illustrate the preceding analyses, one can take the generalized irreversible Brayton heat pump cycle with variable-temperature heat reservoirs (air as the working fluid) as a numerical example. In the calculations, it is set that $T_{Hin} = 290.0$ K, $T_{Lin} = 268.0$ K, $C_v = 0.7165$ kJ/(kg K), $k = 1.4$, $C_p = 1.0031$ kJ/(kg K), $C_H = C_L = 1.2$ kW/K, $C_i = 0.005$ kW/K, $\phi = 1.01$, $U_T = 5$ kW/K and $\dot{m} = 1.1165$ kg/s. If there are no special explanations, the parameters are set as above. The working fluid temperature T_6 is a variable, and its reasonable value is greater than T_{Hin}. The calculations illustrate that the values of x and y are always in their ranges. The dimensionless heating load is defined as $q_H = Q_H/(\dot{m}T_{Lin}C_v)$.

The COP β versus the dimensionless heating load q_H and the hot-side heat conductance distribution u_{H2} of an endoreversible variable-temperature heat reservoir Brayton heat pump cycle with $C_i = 0$ kW/K and $\phi = 1.0$ is shown in Figure 4. Figure 4 indicates that the relation between q_H and β is monotonically decreasing function for the fixed u_{H2}. The COP versus the dimensionless heating load q_H and the hot-side heat conductance distribution u_{H2} of an irreversible variable-temperature heat reservoir Brayton heat pump cycle with $C_i = 0.005$ kW/K and $\phi = 1.01$ is shown in Figure 5. Figure 5 indicates that the curve of the COP versus the dimensionless heating load q_H is a parabolic-like one for the fixed u_{H2}. There exists an optimal heat conductance distribution ($u_{H2\ \mathrm{opt},\ \beta}$) which leads to the optimal COP ($\beta_{\mathrm{opt,u}}$). Numerical calculation shows that the optimal COP ($\beta_{\mathrm{opt,u}}$) and the optimal dimensionless heating load ($q_{H\ \mathrm{opt,u}}$) correspond to the same optimal hot-side heat conductance distribution $u_{H2\ \mathrm{opt}}$ for a larger dimensionless heating load. For Otto, Diesel and Atkinson heat pump cycles, the three-dimensional diagram characteristics among COP versus dimensionless heating load and heat conductance distribution

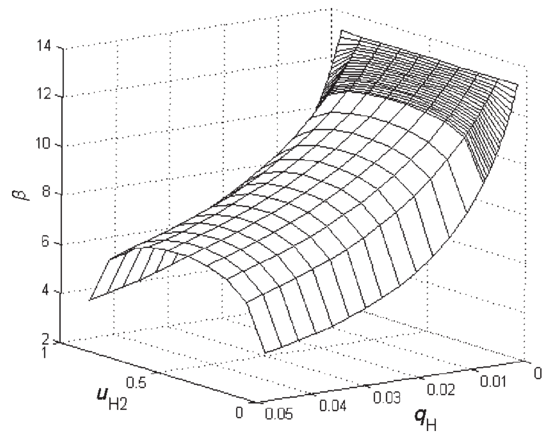

Figure 4. *COP versus the dimensionless heating load and heat conductance distribution of an endoreversible Brayton heat pump cycle.*

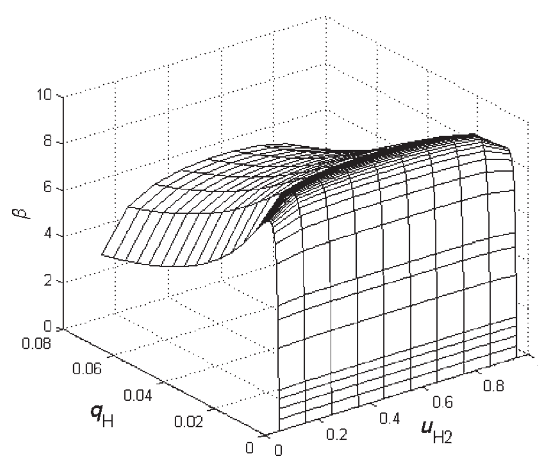

Figure 5. *COP versus the dimensionless heating load and heat conductance distribution of an irreversible Brayton heat pump cycle.*

are similar to those shown in Figures 4 and 5. The three-dimensional diagrams among the COP β and heat conductance distributions (u_{H1} and u_{H2}) of an irreversible variable-temperature heat reservoir Dual heat pump cycle with $q_H = 0.05$, $C_i = 0.005$ kW/K and $\phi = 1.01$ is shown in Figure 6. Figure 6 indicates that there exists a pair of u_{H1} near zero and u_{H2} near 0.5, which lead to the optimal COP. In this case, Dual heat pump cycle becomes Diesel heat pump cycle. The three-dimensional diagrams among the COP β and heat conductance distributions (u_{L1} and u_{L2}) of an irreversible variable-temperature heat reservoir Miller heat pump cycle with $q_H = 0.05$, $C_i = 0.005$ kW/K and $\phi = 1.01$ is shown in Figure 7. Figure 7 indicates that there exists a pair of u_{L1} near 0.5 and u_{L2} near zero, which lead to the optimal COP. In this case, the Miller heat pump cycle becomes the Atkinson heat pump cycle.

Figures 8–10 show the optimal heat conductance distributions versus COP for Brayton, Otto, Diesel, Atkinson, Dual and Miller heat pump cycles. One can see that for a larger COP and a fixed total heat exchanger inventory $U_T = 5$ kW/K, the optimal hot-side heat conductance distribution $u_{H2opt,\, q_H}$ is 0.5 for Brayton and Otto heat pump cycles; $u_{H2opt,\, q_H}$ always approaches

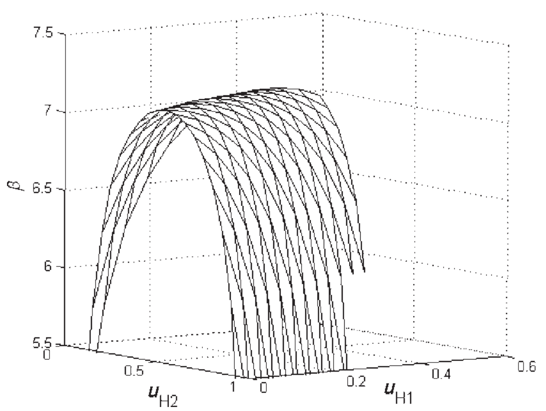

Figure 6. *COP versus distributions of heat conductance of an irreversible Dual heat pump cycle.*

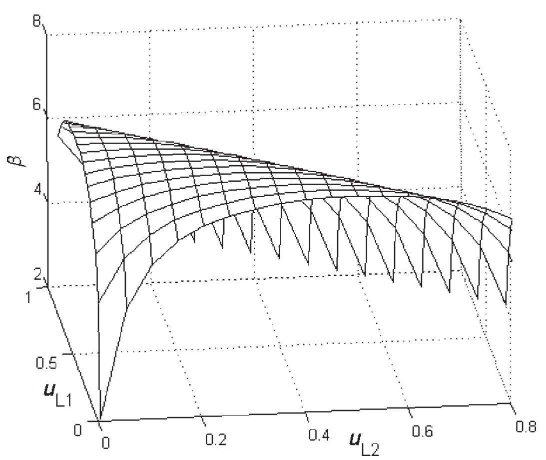

Figure 7. *COP versus distributions of heat conductance of an irreversible Miller heat pump cycle.*

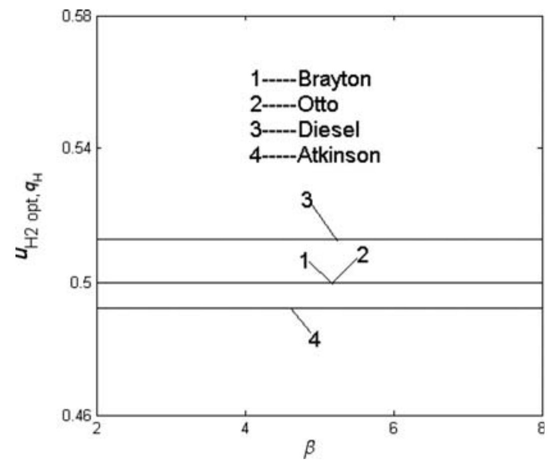

Figure 8. u_{H2opt,q_H} *versus COP for the irreversible Brayton, Otto, Diesel and Atkinson heat pump cycles.*

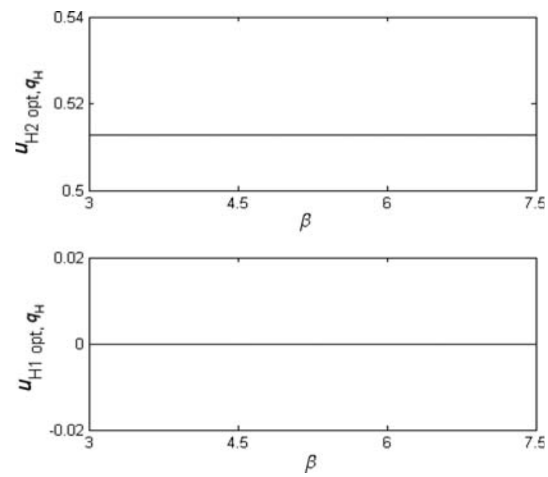

Figure 9. u_{H1opt,q_H} *and* $u_{H2\ opt,q_H}$ *versus* β *for an irreversible Dual heat pump cycle.*

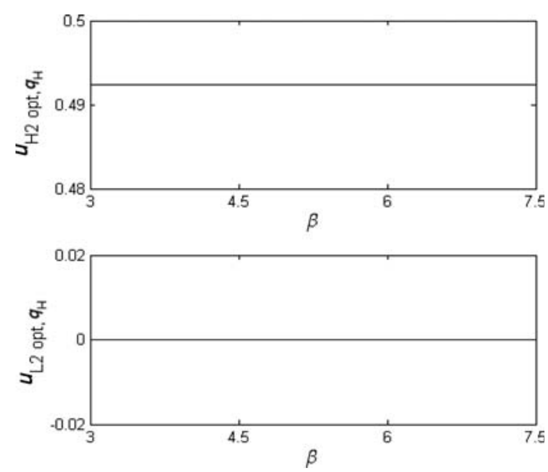

Figure 10. u_{H2opt,q_H} *and* u_{L2opt,q_H} *versus* β *for an irreversible Miller heat pump cycle.*

0.5 for Diesel and Atkinson heat pump cycles; $u_{H1opt,\,q_H}$ approaches zero for the Dual heat pump cycle; $u_{L2opt,\,q_H}$ approaches zero for the Miller heat pump cycle. Moreover, comparing Figures 9 and 10 with Figure 8, one can find that the optimal hot-side heat conductance distributions $u_{H2opt,\,q_H}$ of Dual and Miller heat pump cycles are the same as the corresponding ones of Diesel and Atkinson heat pump cycles for a larger COP, respectively. In this case, Dual and Miller heat pump cycles become Diesel and Atkinson heat pump cycles, respectively.

5.2 Optimal heating load and COP

Figure 11 shows the optimal dimensionless heating load versus COP characteristic for generalized irreversible Brayton, Otto, Diesel, Atkinson, Dual and Miller heat pump cycles with variable-temperature heat reservoirs. Figure 11 indicates that the optimal dimensionless heating load decreases with the increase in COP, but the curve of the COP versus heating load is a parabolic-like one as shown in Figure 3. This is because there exist two heating loads for the fixed COP for the irreversible heat pump cycles with variable-temperature heat reservoirs, and the larger heating load is always chosen as the optimization objective when proceeding to heat conductance optimization. Dual and Diesel, Miller and Atkinson heat pump cycles have the same optimal dimensionless heating load versus COP characteristics. For the fixed COP, Brayton heat pump cycle has the maximum optimal dimensionless heating load among the six heat pump cycles, and Otto heat pump cycle has the minimum.

Figures 12 and 13 show the effects of heat leakage C_i and internal irreversibility ϕ on the optimal COP versus the dimensionless heating load characteristics of an irreversible Brayton heat pump cycle with variable-temperature heat reservoirs, respectively. They indicate that the optimal COP decreases with the increase in the dimensionless heating load when $C_i = 0$; the curve of the optimal COP versus the dimensionless heating load is a parabolic-like one when $C_i \neq 0$. For the fixed dimensionless heating load, the optimal COPs

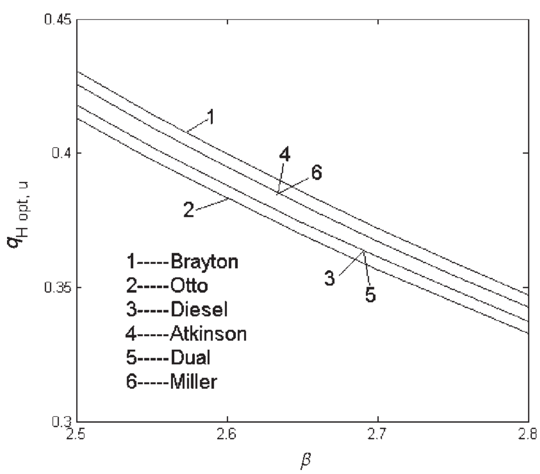

Figure 11. *$q_{H\ opt,u} - \beta$ characteristics for six heat pump cycles.*

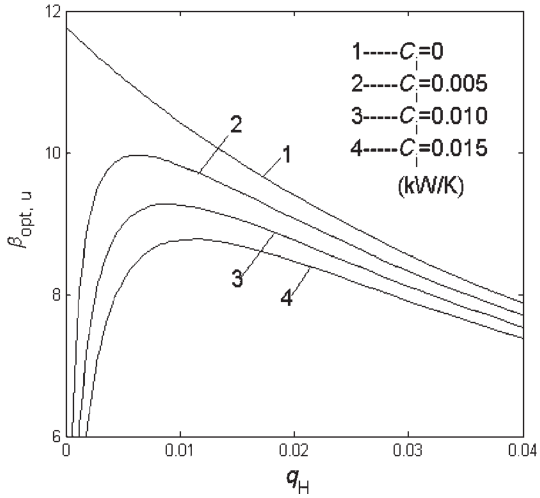

Figure 12. *Effect of C_i on $\beta_{opt,u} - q_H$ characteristic for a Brayton heat pump cycle.*

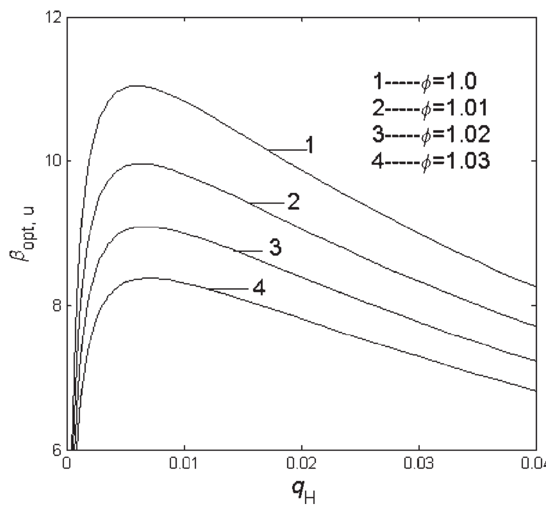

Figure 13. *Effect of ϕ on $\beta_{opt,u} - q_H$ characteristic for a Brayton heat pump cycle.*

($\beta_{opt,\ u}$) decreases with the increases in the heat leakage C_i and internal irreversibility ϕ.

Figures 14 and 15 show the effects of heat leakage C_i and internal irreversibility ϕ on the optimal dimensionless heating load versus COP characteristics of an irreversible Brayton heat pump cycle with variable-temperature heat reservoirs, respectively. It can be seen that the optimal dimensionless heating loads ($q_{H\ opt,u}$) decrease with the increases in the heat leakage C_i and internal irreversibility ϕ for the fixed COP.

Figure 16 shows the effect of total heat exchanger inventory U_T on the optimal dimensionless heating load versus COP characteristic of an irreversible Brayton heat pump cycle with variable-temperature heat reservoirs. It indicates that the optimal dimensionless heating load ($q_{H\ opt,u}$) increases with the increase in total heat exchanger inventory U_T for the fixed COP, but the increment decreases gradually. Figure 17 shows the

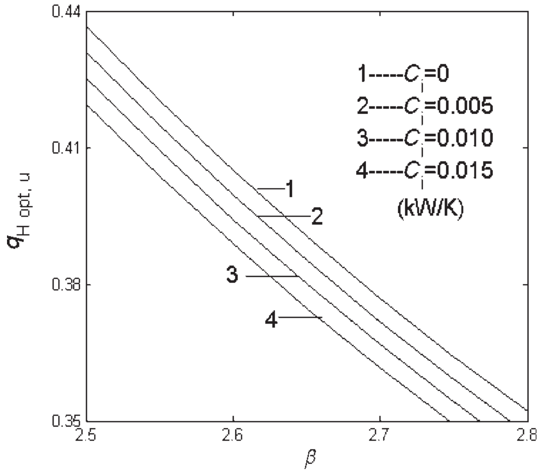

Figure 14. *Effect of C_i on $q_{H\ opt,u} - \beta$ characteristic for a Brayton heat pump cycle.*

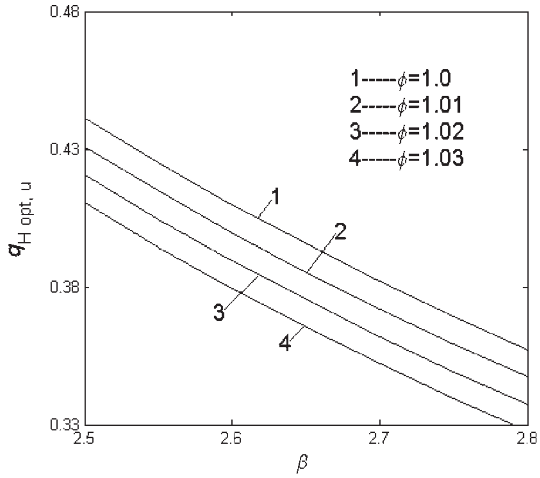

Figure 15. *Effect of ϕ on $q_{H\ opt,u} - \beta$ characteristic for a Brayton heat pump cycle.*

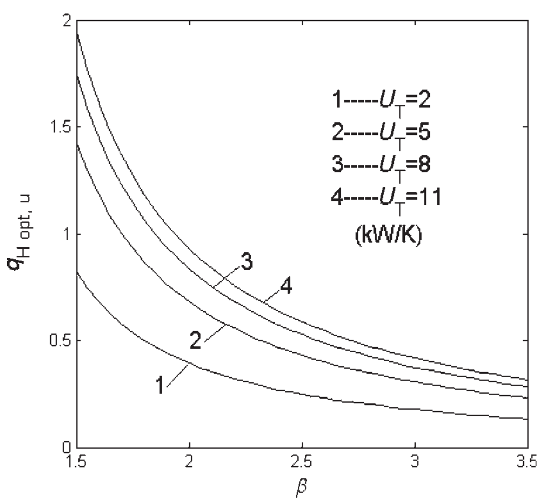

Figure 16. *Effect of U_T on $q_{H\ opt,u} - \beta$ characteristic for a Brayton heat pump cycle.*

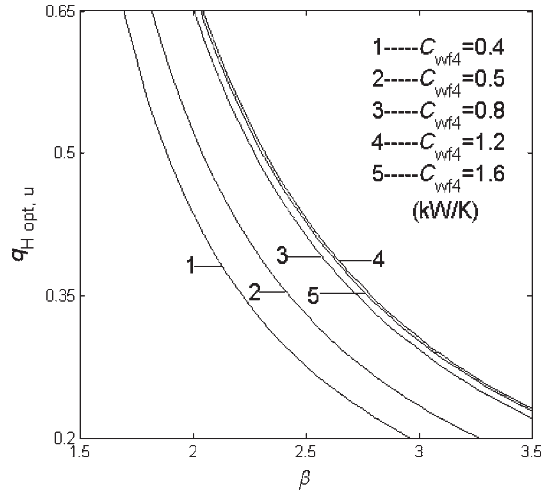

Figure 17. *Effect of C_{wf4} on $q_{H\ opt,u} - \beta$ characteristic for a Brayton heat pump cycle.*

effect of thermal capacity rate of the working fluid C_{wf4} on the optimal dimensionless heating load versus COP characteristic of the irreversible variable-temperature heat reservoir Brayton heat pump cycle with $C_H = C_L = 1.2$ kW/K. It indicates that the optimal dimensionless heating load increases with the increase in thermal capacity rate of the working fluid C_{wf4} when C_{wf4} is lower than C_H and C_L; the optimal dimensionless heating load decreases with the increase in thermal capacity rate of the working fluid C_{wf4} when C_{wf4} is greater than C_H and C_L.

For Otto, Diesel, Atkinson, Dual and Miller heat pump cycles, the effects of heat leakage, internal irreversibility, total heat exchanger inventory and thermal capacity rate of the working fluid on the optimal performance are similar to those shown in Figures 12–17.

5.3 Optimal thermal capacity rate matching between the working fluid and heat reservoirs

Figure 18 shows a three-dimensional diagrams among the dimensionless heating load q_H, thermal capacity rate matching (C_{wf4}/C_L) between the working fluid and heat reservoirs and heat conductance distribution u_{H2} of an irreversible variable-temperature heat reservoir Brayton heat pump cycle with $\beta = 3$, $C_i = 0.005$ kW/K and $\phi = 1.01$. It indicates that the curve of the dimensionless heating load versus the thermal capacity rate matching (C_{wf4}/C_L) between the working fluid and heat reservoirs is a parabolic-like one for the fixed heat conductance distribution u_{H2}. There exist an optimal thermal capacity rate matching ($(C_{wf4}/C_L)_{opt}$) between the working fluid and heat reservoirs and an optimal heat conductance distribution ($u_{H2\ opt}$) which lead to the double maximum dimensionless heating load.

Figures 19 and 20 show the effects of heat leakage C_i and internal irreversibility ϕ on the optimal dimensionless heating load versus the thermal capacity rate matching (C_{wf4}/C_L) between the working fluid and heat reservoirs characteristics of an irreversible Brayton heat pump cycle with variable-temperature heat

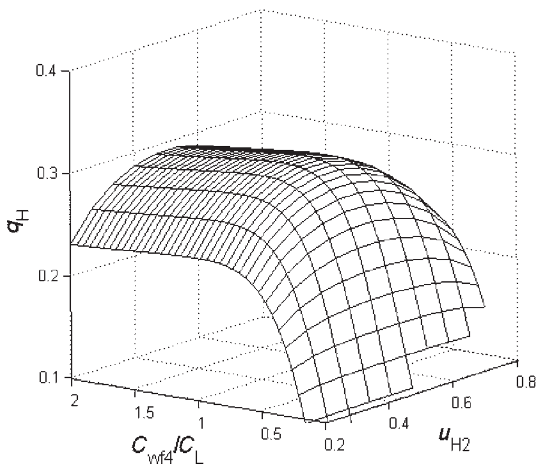

Figure 18. *q_H versus C_{wf4}/C_L and u_{H2} for an irreversible Brayton heat pump cycle.*

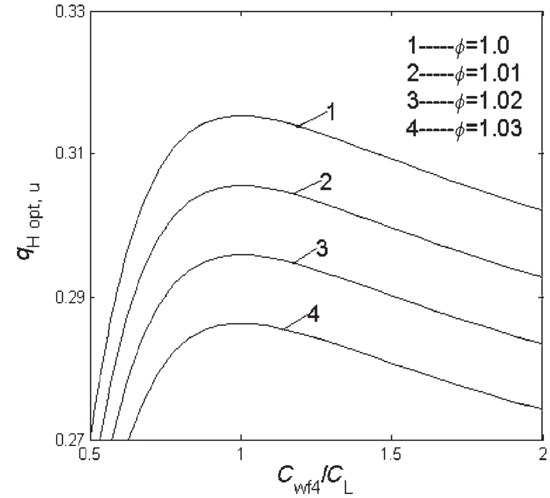

Figure 20. *Effect of ϕ on $q_{H\ opt,u} - C_{wf4}/C_L$ characteristic for a Brayton heat pump cycle.*

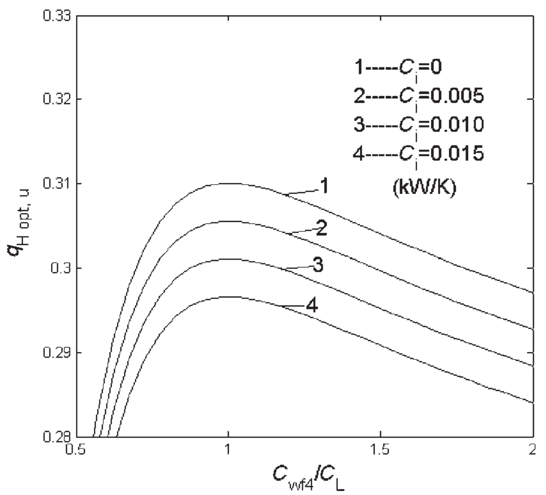

Figure 19. *Effect of C_i on $q_{H\ opt,u} - C_{wf4}/C_L$ characteristic for a Brayton heat pump cycle.*

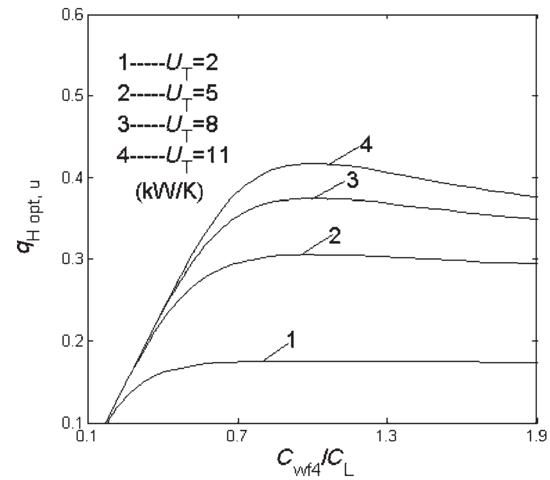

Figure 21. *Effect of U_T on $q_{H\ opt,u} - C_{wf4}/C_L$ characteristic for a Brayton heat pump cycle.*

reservoirs, respectively. It can be seen that the optimal dimensionless heating loads ($q_{H\ opt,u}$) decreases with the increases in the heat leakage C_i and internal irreversibility ϕ for the fixed thermal capacity rate matching C_{wf4}/C_L between the working fluid and heat reservoirs; the heat leakage C_i and internal irreversibility ϕ have little effects on the corresponding optimal thermal capacity rate matching ($(C_{wf4}/C_L)_{opt}$) between the working fluid and heat reservoirs.

Figure 21 shows the effect of total heat exchanger inventory U_T on the optimal dimensionless heating load versus thermal capacity rate matching between the working fluid and heat reservoirs characteristic of an irreversible Brayton heat pump cycle with variable-temperature heat reservoirs. It indicates that the optimal dimensionless heating load ($q_{H\ opt,u}$) increases with the increase in total heat exchanger inventory U_T for the fixed thermal capacity rate matching between the working fluid and heat reservoirs, but the increment decreases gradually; the optimal thermal capacity rate matching ($(C_{wf4}/C_L)_{opt}$) between the working fluid and heat reservoirs increases with the increase in the total heat exchanger inventory U_T. Particularly, the optimal thermal capacity rate matching between the working fluid and heat reservoirs becomes $(C_{wf4}/C_L)_{opt} = 1$ when $C_H/C_L = 1.0$.

For Otto, Diesel, Atkinson, Dual and Miller heat pump cycles, the effects of heat leakage, internal irreversibility, total heat exchanger inventory and thermal capacity rate ratio of the heat reservoirs on the optimal performance are similar to those shown in Figures 19–21.

6 CONCLUSION

The heating load and COP optimizations are carried out in order to investigate the optimal performance of a class of generalized irreversible universal steady-flow heat pump cycle model consisting of two heat-absorbing branches, two heat-releasing branches and two adiabatic branches with variable-temperature heat reservoirs by using the theory of finite-time thermodynamics. Heat resistance, heat leakage and internal irreversibility are considered. Expressions for heating load and COP are derived and used to discuss the optimal performance of the universal heat pump cycle. Numerical examples show that the optimal hot-side heat conductance distribution u_{H2opt,q_H} is 0.5 for Brayton and Otto heat pump cycles; u_{H2opt,q_H} always approaches 0.5 for Diesel, Atkinson, Dual and Miller heat pump cycles; moreover, u_{H2opt,q_H} for Dual and Miller heat pump cycles are the same as the corresponding ones of Diesel and Atkinson heat pump cycles for a larger COP, respectively. There exist an optimal heat conductance distribution and an optimal thermal capacity rate matching between the working fluid and heat reservoirs which lead to the double maximum heating load. Moreover, the effects of heat leakage, internal irreversibility, total heat exchanger inventory and thermal capacity rate of the working fluid on the optimal heating load and optimal thermal capacity rate matching between the working fluid and heat reservoirs are discussed. The results obtained herein include the optimal performance of endoreversible and irreversible, constant- and variable-temperature heat reservoir Brayton, Otto, Diesel, Atkinson, Dual, Miller and Carnot heat pump cycles. This paper first considered heat leakage in the universal heat pump cycle with variable-temperature heat reservoirs. Heat resistance, heat leakage and internal irreversibility are simultaneously integrated in the cycle model which is closer to practical heat pumps. The results obtained in this paper may provide some theoretical guidelines for parameter designs and performance optimizations of practical heat pumps.

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